

```
#!/usr/bin/env python3 """ ShelbyFirst: Full Outcomes Improvement Pathway Run: python shelby-
first_outcomes_chart.py Output: src/data/processed/shelbyfirst_outcomes_chart.png """
```

```
import matplotlib matplotlib.use('Agg') import matplotlib.pyplot as plt from matplotlib.patches import
Rectangle from matplotlib.lines import Line2D import numpy as np import os import pandas as pd from
sklearn.linear_model import Ridge from sklearn.preprocessing import StandardScaler from sklearn.pipeline
import Pipeline
```

— Palette —

```
NAV = '#1b2a6b' # outer bg, header, table col-header TBL = '#2e4296' # table section separator WHT =
'#####' EVN = '#eef1fa' # alternating table row GRY = '#dee3ef' # grid / cell borders RED = '#c0392b' # "Now"
values GRN = '#1a7a35' # "Yr 3" values BLU = '#1a4db5' # "Peer" values DRK = '#1a1a2e' # normal row
text SUB = '#8898cc' # subtitle / footer text
```

— Model-driven projections —

Follows the ShelbyFirst causal pathway (PDF slide 8):

Programs → direct metric improvements → IGS rises as a consequence

”ShelbyFirst influences IGS indirectly through better mobility, healthier

food access, stronger preventive care participation, and deeper community

engagement.” — PDF slide 8

Two-layer model for economic metrics:

**Layer 1 — Ridge Regression (year-only, no IGS) on 2017-2024 panel
= historical trend / counterfactual without ShelbyFirst**

Layer 2 — ShelbyFirst gap-closing boost layered on top

boost = (peer_benchmark – baseline) × effectiveness × uptake

Health / access metrics: gap-closing model only (no tract-level panel data)

Program uptake by phase (PDF slide 6 — Implementation Plan):

Year 1 Pilot = 15% founding households, initial ride utilization

Year 2 Expand = 35% growing membership, expanded programs

Year 3 Scale = 60% full rollout across low-IGS tracts

```
_dir = os.path.dirname(os.path.abspath(file)) _data = os.path.join(_dir, 'data', 'processed')
```

```

_panel = pd.read_parquet(os.path.join(_data, 'enrichment_tract_year_panel.parquet')) _shelby =
_panel[_panel['geoid'].astype(str).str.startswith('47157')].copy() _shelby['income_k'] = _shelby['median_household_income']
/ 1000

_YEARS = [2024, 2025, 2026, 2027] UPTAKE = [0.0, 0.15, 0.35, 0.60] # fraction of 96-tract population
engaged

```

Peer benchmarks (PDF slide 4 / slide 5)

```

_PEER = { 'labor_force': 66.9, 'poverty': 5.5, 'income': 119.0, 'insured': 94.0, 'no_veh': 4.2, 'dental': 32.1,
'obesity': 32.5, 'diabetes': 11.3, }

```

Effectiveness: fraction of peer gap closeable per unit of uptake.

Grounded in ShelbyFirst's three pillars (PDF slide 7):

Move → affordable rides address transportation friction (PDF slide 5)

Nourish → community garden targets obesity and diabetes risk factors

Connect → app + ShelbyCares referrals drive enrollment and care access

```

EFFECT = { 'labor_force': 0.22, # Connect: health improvement → employability 'poverty': 0.15, # Cross-
pillar: employment + economic participation 'income': 0.10, # Very conservative — structural income change
is slow 'insured': 0.40, # Connect: active enrollment navigation + ShelbyCares 'no_veh': 0.20, # Move:
provides alternative mobility, not car ownership 'dental': 0.35, # Move: transport is the primary dental gap
driver 'obesity': 0.22, # Nourish: garden + physical activity programs 'diabetes': 0.18, # Nourish: diet +
nutrition access improvement }

```

```

def _ridge_trend(df, col, scale=1.0): """Year-only Ridge: counterfactual trend without ShelbyFirst inter-
vention.""" d = df[['year', col]].dropna() pipe = Pipeline([('sc', StandardScaler()), ('r', Ridge(alpha=1.0))])
pipe.fit(d[['year']].values, d[col].values) return (pipe.predict(np.array([[yr] for yr in _YEARS])) * scale).tolist()

```

```

def _gap_boost(baseline, peer, pillar, direction): """ShelbyFirst gap-closing: baseline ± gap × effectiveness
× uptake per year.""" gap = abs(peer - baseline) eff = EFFECT[pillar] return [baseline + direction * gap * eff
* u for u in UPTAKE]

```

```

def _econ(df, col, scale, peer, pillar, direction): """Layer 1 (Ridge trend) + Layer 2 (ShelbyFirst boost)."""
trend = _ridge_trend(df, col, scale) gap = abs(peer - trend[0]) eff = EFFECT[pillar] result = [trend[0]] for i in
range(1, 4): result.append(trend[i] + direction * gap * eff * UPTAKE[i]) return result

```

```

_lfpr_v = _econ(_shelby, 'lfpr_16p', 100, _PEER['labor_force'], 'labor_force', 1) _pov_v = _econ(_shelby,
'poverty_rate', 100, _PEER['poverty'], 'poverty', -1) _inc_v = _econ(_shelby, 'income_k', 1, _PEER['income'],
'income', 1)

```

Insured baseline from cluster healthcare data

```

_cs = pd.read_parquet(os.path.join(_data, 'cluster_shortlist_with_healthcare.parquet')) _r77 = _cs[_cs['rank']
== 11] _ins0 = (1 - float(_r77.iloc[0]['uninsured_rate'])) * 100

```

```
if len(_r77) and not pd.isna(_r77.iloc[0]['uninsured_rate']) else 82.0
```

```
_ins_v = _gap_boost(_ins0, _PEER['insured'], 'insured', 1) _noveh_v = _gap_boost(13.8, _PEER['no_veh'],
'no_veh', -1) _dental_v = _gap_boost(55.6, _PEER['dental'], 'dental', -1) _obes_v = _gap_boost(45.2,
_PEEER['obesity'], 'obesity', -1) _diab_v = _gap_boost(21.6, _PEER['diabetes'], 'diabetes', -1)
```

IGS: consequence of program improvements, shown as the planning target arc

```
_IGS = [35.8, 36.8, 38.4, 40.5]
```

— Metric definitions —

(table_label, direction, [baseline,yr1,yr2,yr3], peer, line_color, legend_label, fmt)

```
METRICS = [ ('IGS Score', 1, _IGS, 55.7, '#4545c0', f'IGS Score ({_IGS[0]:.1f} → {_IGS[-1]:.1f})', '{:.1f}'),
('Labor Force %', 1, _lfpr_v, _PEER['labor_force'], '#1a8c5a', f'Labor Force Part. ({_lfpr_v[0]:.1f}% → {_lfpr_v[-1]:.1f}%)', '{:.1f}%'),
('Insured %', 1, _ins_v, _PEER['insured'], '#c9a000', f'Insured Rate ({_ins_v[0]:.1f}% → {_ins_v[-1]:.1f}%)', '{:.1f}%'),
('Med. Income', 1, _inc_v, _PEER['income'], '#e06000', f'Med. Income (${_inc_v[0]:.1f}k → ${_inc_v[-1]:.1f}k)', '${:.1f}k'),
('Poverty %', -1, _pov_v, _PEER['poverty'], '#8b1515', f'Poverty Rate ({_pov_v[0]:.1f}% → {_pov_v[-1]:.1f}%)', '{:.1f}%'),
('No-Veh HH %', -1, _noveh_v, _PEER['no_veh'], '#cc1515', f'No-Vehicle HH ({_noveh_v[0]:.1f}% → {_noveh_v[-1]:.1f}%)', '{:.1f}%'),
('Dental Gap %', -1, _dental_v, _PEER['dental'], '#1a8cd4', f'Dental Visit Gap ({_dental_v[0]:.1f}% → {_dental_v[-1]:.1f}%)', '{:.1f}%'),
('Obesity %', -1, _obes_v, _PEER['obesity'], '#8bc820', f'Obesity Rate ({_obes_v[0]:.1f}% → {_obes_v[-1]:.1f}%)', '{:.1f}%'),
('Diabetes %', -1, _diab_v, _PEER['diabetes'], '#30a040', f'Diabetes Rate ({_diab_v[0]:.1f}% → {_diab_v[-1]:.1f}%)', '{:.1f}%'), ]
```

```
X = [0, 1, 2, 3] X_LABELS = ['Baseline\n2024', 'Year 1 — Pilot\n2025', 'Year 2 — Expand\n2026', 'Year 3 — Scale\n2027', ]
```

```
def improvement_index(direction, values): """All metrics converted so that higher index = improvement."""
base = values[0] return [v / base * 100 for v in values] if direction == 1
else [base / v * 100 for v in values]
```

— Figure —

Tall enough so the legend strip (22%-8%) never overlaps the chart (22%-88%).

```
fig = plt.figure(figsize=(20, 12), dpi=130, facecolor=NAV)
```

HEADER

```
fig.text(0.015, 0.965, 'ShelbyFirst: Full Outcomes Improvement Pathway', color=WHT, fontsize=21,
fontweight='bold', va='top') fig.text(0.015, 0.922, 'Economic + health metrics · ShelbyFirst gap-closing
model (Move · Nourish · Connect pillars) · 'Uptake: 15% Pilot → 35% Expand → 60% Scale · 96 tracts ·
pop. 273K', color=SUB, fontsize=10, va='top')
```

CHART — left panel [left, bottom, width, height] all in figure fractions

Now starts at 0.045 (no rotated label needed on the left any more)

```
CHART_L, CHART_B, CHART_W, CHART_H = 0.048, 0.23, 0.500, 0.655 ax = fig.add_axes([CHART_L,
CHART_B, CHART_W, CHART_H], facecolor=WHT)
ax.set_axisbelow(True) ax.yaxis.grid(True, color=GRY, linewidth=0.8) ax.set_xlim(-0.35, 3.35) ax.set_ylim(94,
147) ax.set_xticks(X) ax.set_xticklabels(X_LABELS, fontsize=11.5, color=WHT, linespacing=1.6,
fontweight='bold') ax.set_yticks(range(95, 150, 5)) ax.set_yticklabels([str(v) for v in range(95, 150, 5)],
fontsize=11, color=WHT, fontweight='bold') ax.tick_params(axis='both', length=0) ax.tick_params(axis='x',
pad=12) ax.tick_params(axis='y', pad=6)
for spine in ['top', 'right', 'left']: ax.spines[spine].set_visible(False) ax.spines['bottom'].set_color(GRY)
```

— Horizontal note just above the chart (no title, just the definition line)

```
ax.text(0.5, 1.008, '100 = 2024 baseline · ↑ higher = better for all 9 metrics', transform=ax.transAxes,
ha='center', va='bottom', fontsize=10.5, color='#aabfee', style='italic')
```

— "Improvement Index" — rotated Y-axis label in figure space (avoids clipping)

```
CHART_CX = CHART_L + CHART_W / 2 # figure-fraction x centre of chart fig.text(0.010, CHART_B
+ CHART_H / 2, 'Improvement Index', ha='center', va='center', rotation=90, fontsize=12.5, color=WHT,
fontweight='bold')
```

Baseline reference line (dashed) — explanation text moved to subtitle above

```
ax.axhline(y=100, color='#8090c0', linewidth=1.3, linestyle='--', zorder=2)
```

Plot lines & collect handles

```
legend_handles = [] for tbl_lbl, direction, vals, peer, color, leg_lbl, fmt in METRICS: idx = improve-  
ment_index(direction, vals) ax.plot(X, idx, color=color, linewidth=2.4, marker='o', markersize=6.5, mark-  
erfacecolor=WHT, markeredgewidth=1.9, zorder=3) legend_handles.append(  
Line2D([0], [0], color=color, linewidth=2.4, marker='o', markersize=6, markerfacecolor=WHT, markeredge-  
color=color, markeredgewidth=1.7, label=leg_lbl) )
```

LEGEND — sits in the navy band BELOW THE TABLE (right panel)

Table left = CHART_L + CHART_W + 0.012 = 0.560, width = 0.425

Table centre x = 0.560 + 0.425/2 = 0.7725

```
TABLE_L = CHART_L + CHART_W + 0.012 TABLE_W = 0.425 TABLE_CX = TABLE_L + TABLE_W / 2 #  
0.7725
```

```
leg = fig.legend( handles=legend_handles, loc='lower center', bbox_to_anchor=(TABLE_CX, 0.022), ncol=2,  
fontsize=10.5, frameon=True, handlelength=2.2, handleheight=1.0, columnspacing=1.8, labelcolor=WHT,  
borderpad=0.9, labelspace=0.8, facecolor='#1e3070', edgecolor='#4a60b0', framealpha=1.0, ) for text in  
leg.get_texts(): text.set_fontweight('semibold')
```

TABLE — right panel

```
ax_t = fig.add_axes([CHART_L + CHART_W + 0.012, CHART_B, 0.425, CHART_H]) ax_t.set_xlim(0, 1)  
ax_t.set_ylim(0, 1) ax_t.axis('off')
```

```
COL_W = [0.310, 0.128, 0.115, 0.115, 0.115, 0.127] COL_X = list(np.cumsum([0.0] + COL_W[:-1])) COL_H  
= ['Metric', 'Now', 'Yr 1', 'Yr 2', 'Yr 3', 'Peer']
```

```
ROW_H_HDR = 0.082 ROW_H_SEC = 0.050 ROW_H_DATA = 0.073
```

```
TABLE_ROWS = [ ('hdr', None), ('sec', '— ECONOMIC METRICS · Ridge year-trend + ShelbyFirst Con-  
nect boost —————'), ('data', METRICS[0]), ('data', METRICS[1]), ('data', METRICS[2]), ('data', MET-  
RICS[3]), ('data', METRICS[4]), ('data', METRICS[5]), ('sec', '— HEALTH METRICS · ShelbyFirst Move  
/ Nourish gap-closing model —————'), ('data', METRICS[6]), ('data', METRICS[7]), ('data', MET-  
RICS[8]), ]
```

```
y = 0.99 data_idx = 0
```

```
for rtype, rcontent in TABLE_ROWS: rh = {'hdr': ROW_H_HDR, 'sec': ROW_H_SEC, 'data': ROW_H_DATA}[rtype]
```

```
if rtype == 'hdr':  
    for i, (cx, cw, ch) in enumerate(zip(COL_X, COL_W, COL_H)):
```

```

        ax_t.add_patch(Rectangle((cx, y - rh), cw, rh,
                                facecolor=NAV, edgecolor='#3a4ea0',
                                linewidth=0.5, zorder=2))
        xt = cx + 0.013 if i == 0 else cx + cw / 2
        ax_t.text(xt, y - rh / 2, ch,
                  ha='left' if i == 0 else 'center', va='center',
                  fontsize=10, color=WHT, fontweight='bold')

elif rtype == 'sec':
    ax_t.add_patch(Rectangle((0, y - rh), 1.0, rh,
                            facecolor=TBL, edgecolor='none', zorder=2))
    ax_t.text(0.013, y - rh / 2, rcontent,
              ha='left', va='center', fontsize=7.8,
              color='#aac8ff', style='italic')

else:
    tbl_lbl, direction, vals, peer, _, _, fmt = rcontent
    bg = EVN if data_idx % 2 == 1 else WHT
    data_idx += 1

    ax_t.add_patch(Rectangle((0, y - rh), 1.0, rh,
                            facecolor=bg, edgecolor=GRY,
                            linewidth=0.3, zorder=1))

    for cx in COL_X[1:]:
        ax_t.plot([cx, cx], [y - rh, y], color=GRY, linewidth=0.4, zorder=2)

    ax_t.text(COL_X[0] + 0.013, y - rh / 2, tbl_lbl,
              ha='left', va='center', fontsize=9.5, color=DRK)

    ax_t.text(COL_X[1] + COL_W[1] / 2, y - rh / 2, fmt.format(vals[0]),
              ha='center', va='center', fontsize=9.5, color=RED, fontweight='bold')

    ax_t.text(COL_X[2] + COL_W[2] / 2, y - rh / 2, fmt.format(vals[1]),
              ha='center', va='center', fontsize=9.5, color=DRK)

    ax_t.text(COL_X[3] + COL_W[3] / 2, y - rh / 2, fmt.format(vals[2]),
              ha='center', va='center', fontsize=9.5, color=DRK)

    ax_t.text(COL_X[4] + COL_W[4] / 2, y - rh / 2, fmt.format(vals[3]),
              ha='center', va='center', fontsize=10, color=GRN, fontweight='bold')

    ax_t.text(COL_X[5] + COL_W[5] / 2, y - rh / 2, fmt.format(peer),
              ha='center', va='center', fontsize=9.5, color=BLU, fontweight='bold')

y -= rh

```

FOOTER

fig.text(0.015, 0.010, 'Model: ShelbyFirst gap-closing — improvement = (peer gap) × program effectiveness × uptake · ' 'Econ metrics add Ridge year-trend as counterfactual baseline · ' 'Effectiveness grounded in Move (rides), Nourish (garden), Connect (app/referrals) pillars · PDF slides 6–8', color=SUB, fontsize=8, va='bottom',)

SAVE

```
_dir = os.path.dirname(os.path.abspath(file)) out = os.path.join(_dir, 'data', 'processed', 'shelby-first_outcomes_chart.png') os.makedirs(os.path.dirname(out), exist_ok=True) plt.savefig(out, dpi=130, bbox_inches='tight', facecolor=NAV, pad_inches=0.2) print(f'Saved: {out}') plt.close()
```